Custom Raspberry Pi 5 Audio & Power Management Pi HAT PCB Review and Revision

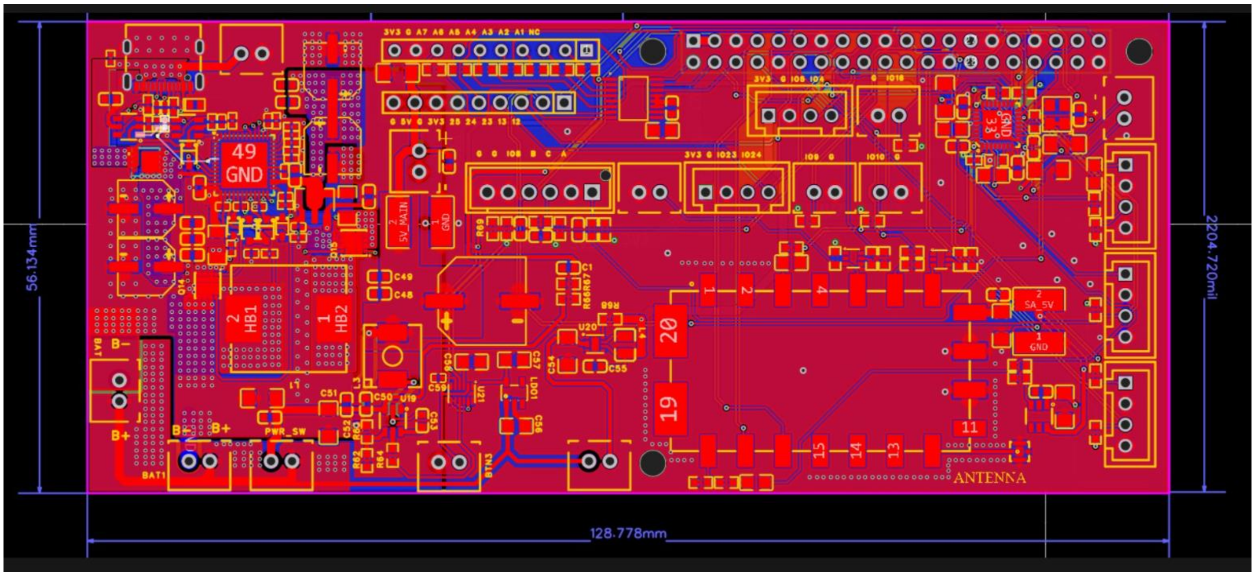
Parameter	Details
PCB Size	**56.134 mm × 128.7mm**
PCB Layers	2-Layer PCB
Design Tool	EasyEDA
Manufacturing Target	JLCPCB Assembly
Main Processor Interface	Raspberry Pi 5 40-Pin Header
Audio Codec	WM8960
Radio Module	SA818 UHF/VHF Transceiver
ADC	ADS7830 8-Channel ADC
Wireless Modules	GPS + LoRa
Power Input	5V–20V DC Input
Battery Support	Multi-cell Li-Ion Battery Management
Power Architecture	USB-C Charging, Buck Converters, Load Switches, Protection Circuitry
Key Features	Audio Routing, Radio Communication, Battery Charging, GPS Tracking, LoRa Connectivity

I recently completed a comprehensive review and redesign of a custom Raspberry Pi 5 Pi HAT PCB developed in EasyEDA for JLCPCB manufacturing and assembly. The project integrated multiple subsystems including a WM8960 audio codec, SA818 radio module, ADS7830 ADC, GPS interface, LoRa connectivity, battery management circuitry, and several power conversion stages. My responsibility was to verify that the PCB implementation accurately matched the design documentation, identify schematic and layout issues, resolve component sourcing concerns, and improve overall system reliability.

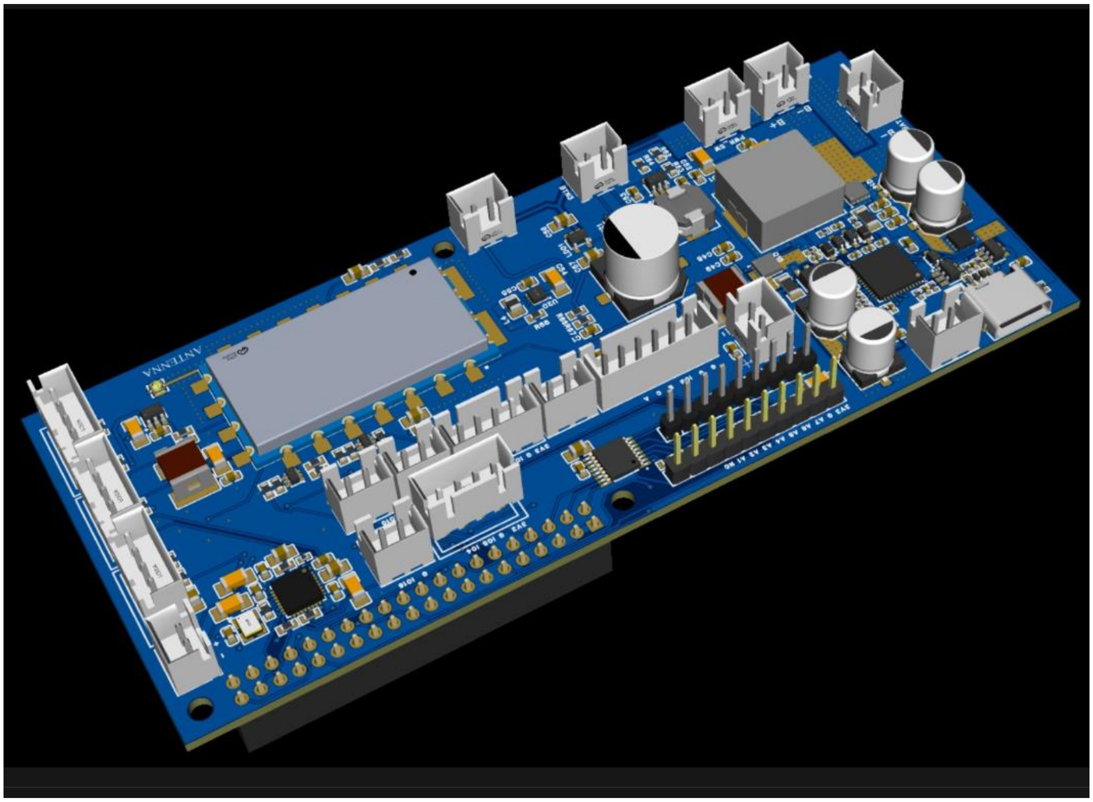
A major focus of the project was analyzing and upgrading the power architecture, which originally had a peak output limitation of approximately 2.4A—insufficient for Raspberry Pi 5 operation alongside connected peripherals. The design review included evaluation of the TPS22918 load switch, battery charging circuitry, buck converters, power-path management, protection circuits, and JLCPCB BOM optimization. The final review ensured improved power delivery capability, manufacturability, and long-term component availability while maintaining compatibility with the audio, radio, and sensing subsystems

PCB

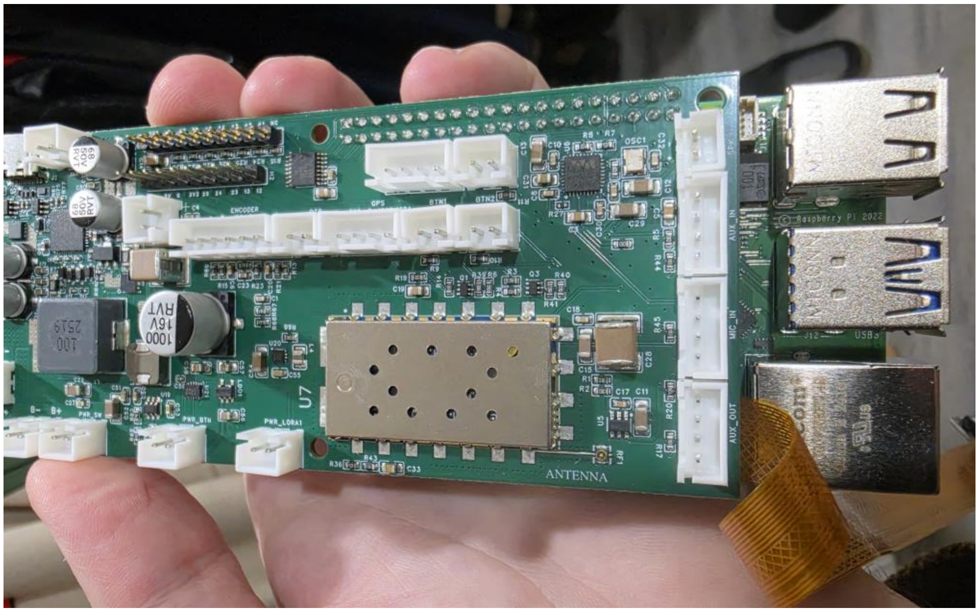
Component placement appears to be functionally grouped, which generally improves signal integrity and routing efficiency.

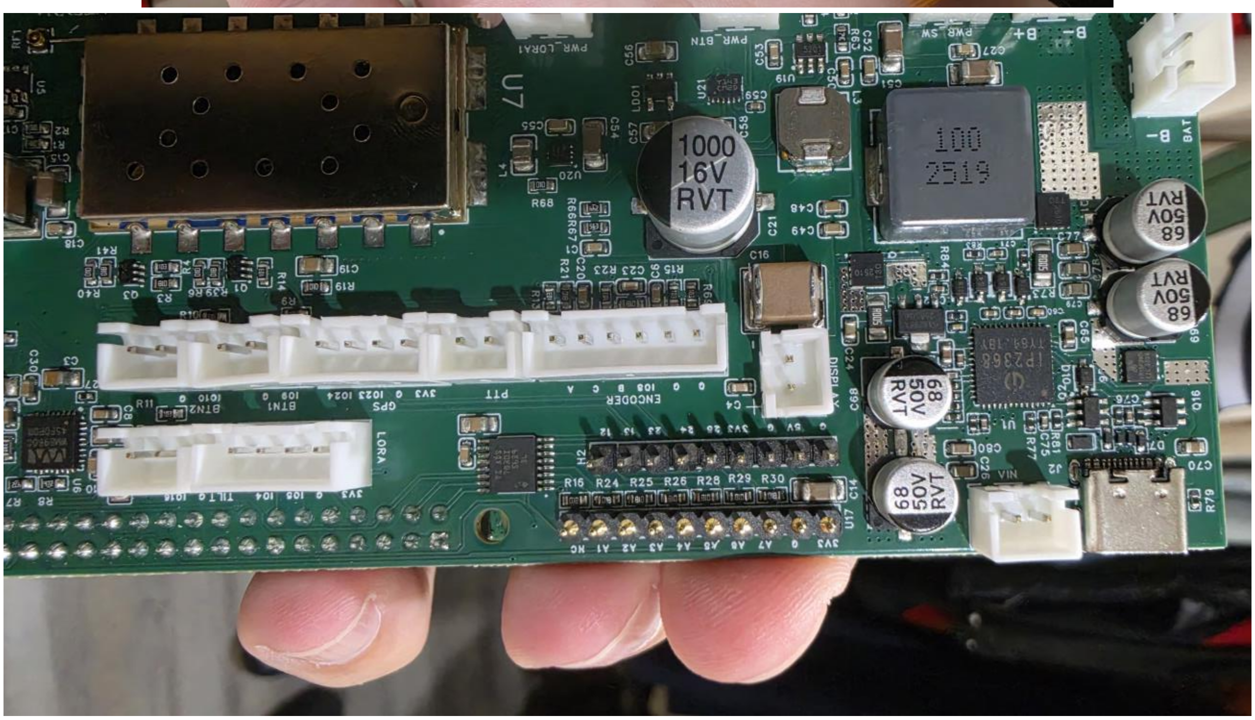


3d PCB



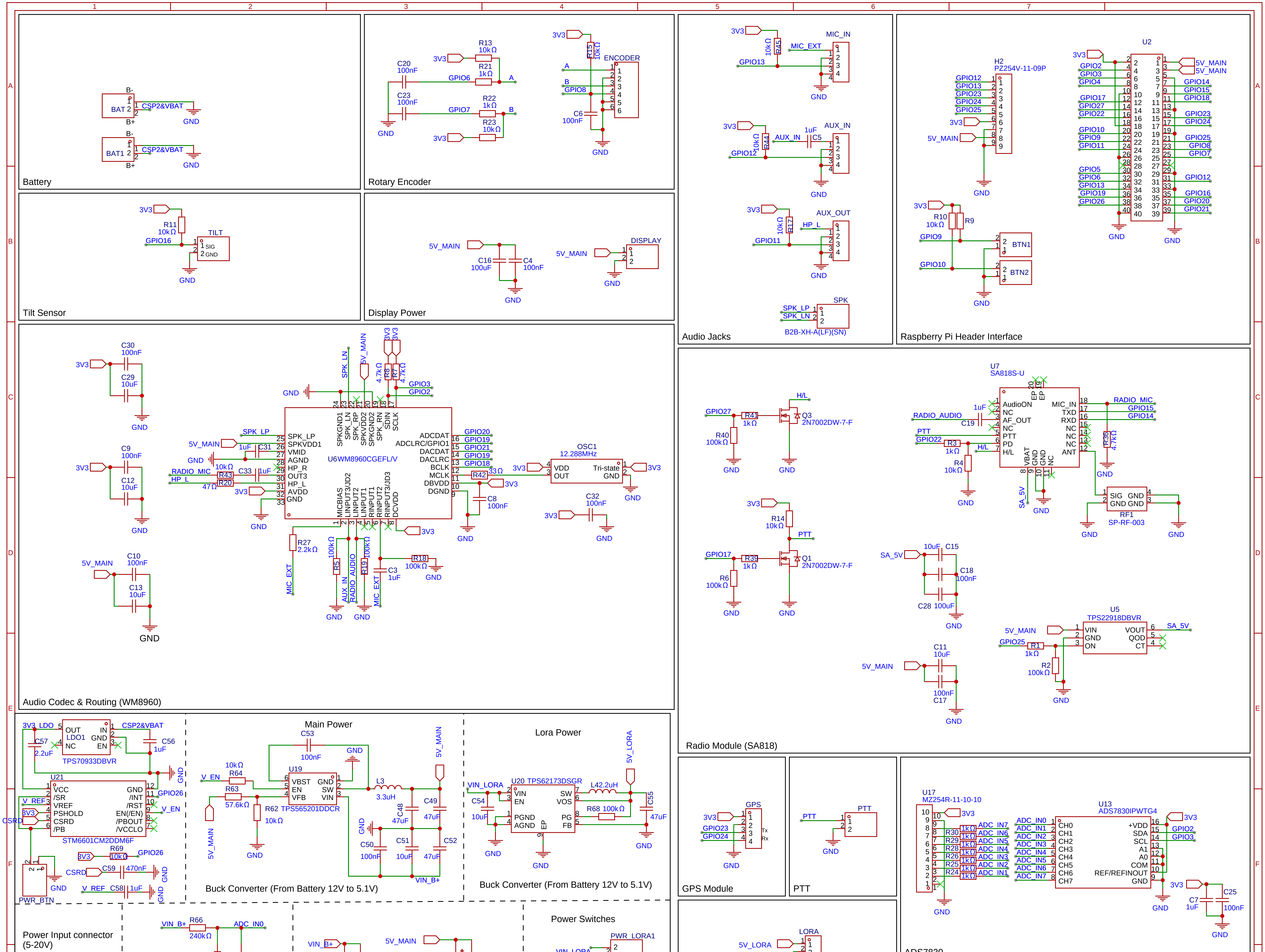
Hardware Tested at Client’s end



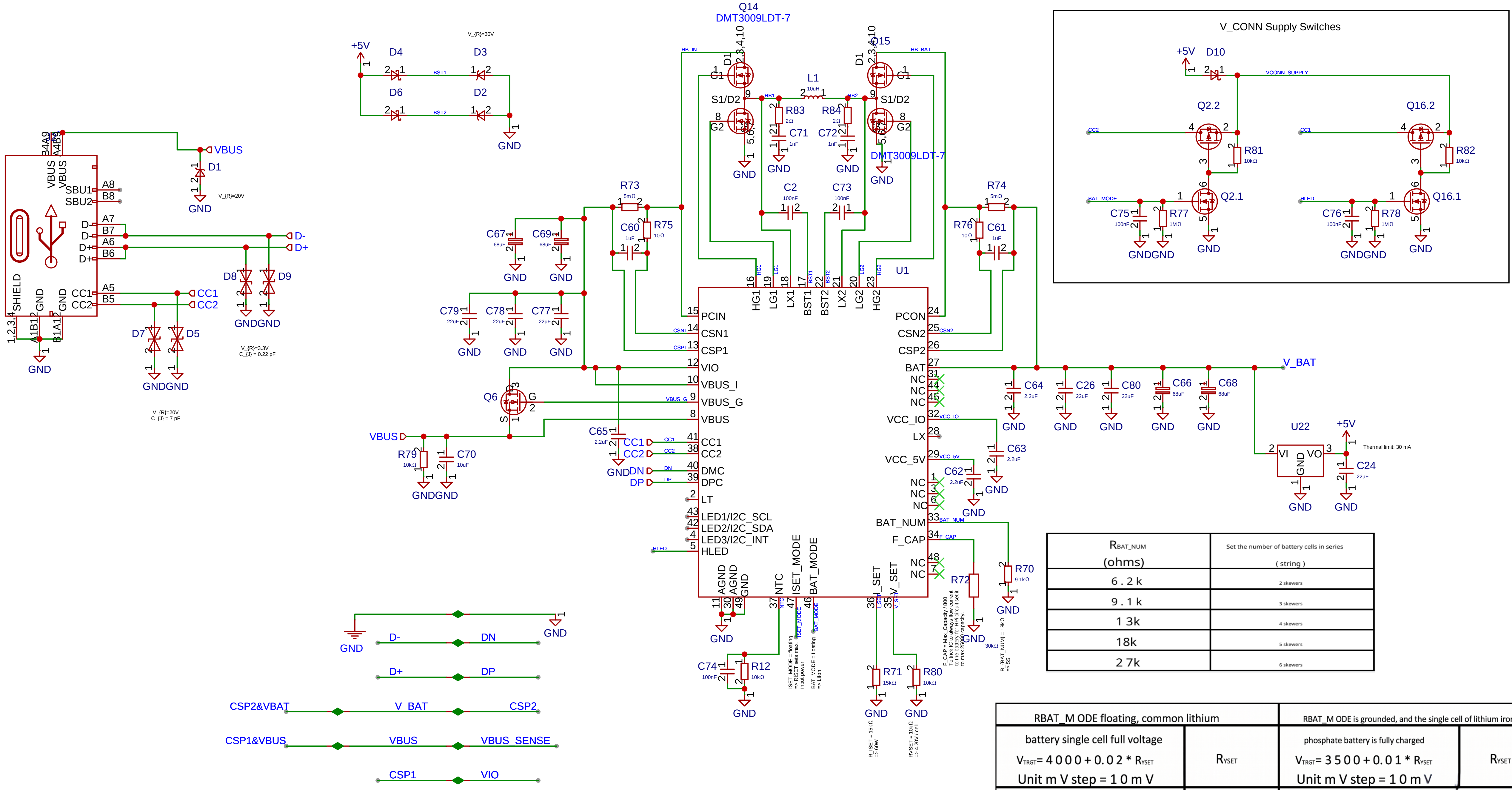


Schematics

The design shows good functional separation between audio, RF, ADC, GPS, LoRa, and power-management sections, making debugging and future revisions easier.



TITLE:	PCB_Jason_atlas_pi_hat_REV2	REV:	1.0
Company:	Your Company	Sheet:	1/1
Date:	2026-03-10	Drawn By:	ahmed.194.abdullah



R_{BAT_NUM} (ohms)	Set the number of battery cells in series (string)
6.2k	2 skewers
9.1k	3 skewers
13k	4 skewers
18k	5 skewers
27k	6 skewers

RBAT_M ODE floating, common lithium		RBAT_M ODE is grounded, and the single cell of lithium iron	
battery single cell full voltage V_{TRGT}= 4000+0.02 * R_{YSET} Unit mV step = 10mV	R_{YSET}	phosphate battery is fully charged V_{TRGT}= 3500+0.01 * R_{YSET} Unit mV step = 10mV	R_{YSET}
4.15V	7.5K	3.55V	7.5K
4.20V	10K	3.60V	10K
4.30V	15K	3.65V	15K
4.35V	17.5K	3.70V	≥20K
4.40V	≥20K		

ISET_M ODE floating RISET sets the constant current maximum input power		ISET_MODE GND RISET sets constant current maximum battery current	
Maximum input power when charging P_{CCIN}=4*R_{ISET} Unit mW step=1W	R_{ISET}	Single battery full voltage I_{CHRG}=0.2*R_{ISET} Unit mA step=100mA	R_{ISET}
20W	5K	1A	5K
30W	7.5K	2A	10K
45W	11.2K	2.5A	12.5K
60W	15K	3A	15K
100W	≥25K	5A	≥25K

Schematic	Jason_atlas_pi_hat	Create at	2026-05-28
		Update at	2026-05-28
Board	Jason_atlas_pi_hat	Page	Battery Charger
Drawn		PCB_Jason_atlas_pi_hat_REV2	
Reviewed			
	Version	Size	Page 2 Total 2
EasyEDA	V1.0	A4	EasyEDA.com